

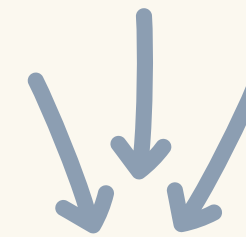
An illustration of a young girl with dark hair in a ponytail, wearing a blue shirt, sitting and reading a red book. The background is light beige with various autumn-themed elements: pinkish-orange leaves and branches, blue plus signs, and blue wavy lines. A large blue speech bubble on the right contains the text "Learning Multiplication".

Learning Multiplication

What is multiplication?



Repeated addition is adding the same number again and again.
It is a simple way to understand multiplication.



Example: $3 + 3 + 3$

is the same as 3 times 3.

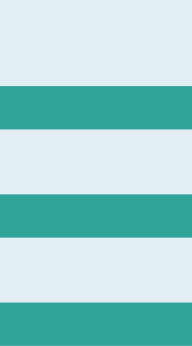
$$3 + 3 + 3 = 9.$$

So, 3 added 3 times equals 9.

LEARN

What is **Multiplication** as **Equal Groups**?

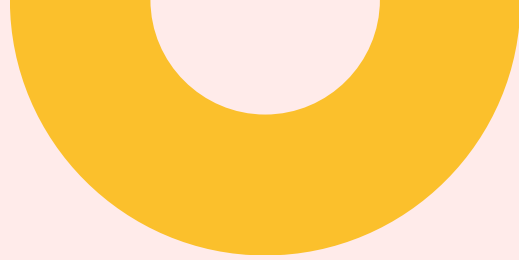
- **Equal groups** mean that each group contains the same number of items.
- **Multiplication** means making more of something. When you multiply, you increase a number in a special way—by joining equal groups.



Terms in Multiplication

The numbers which are multiplied are called the **factors**.

The number that results from multiplying the two **factors** is called the **product**.

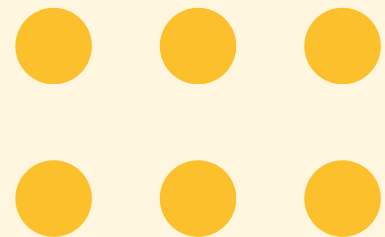

$$\begin{array}{ccc} 5 & \times & 4 & = & 20 \\ \underbrace{\hspace{1.5cm}} & & & & | \\ \text{Factors} & & & & \text{Product} \end{array}$$

Fact Families

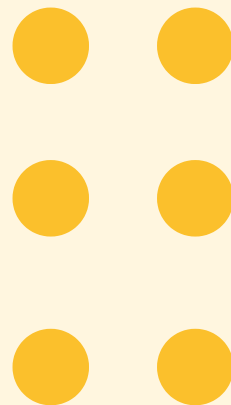
A set of **four** related multiplication and division facts that use the same three numbers.

For example: The fact family for 3, 2, and 6 is a set of two multiplication and two division facts.

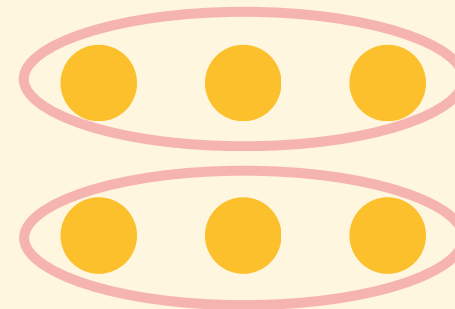
$$2 \times 3 = 6$$



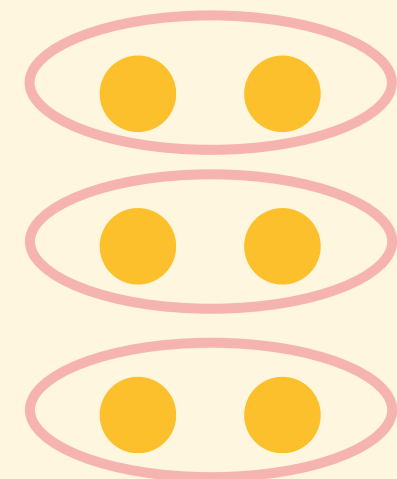
$$3 \times 2 = 6$$



$$6 \div 2 = 3$$

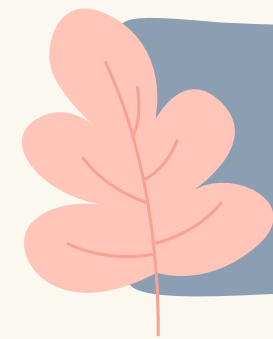


$$6 \div 3 = 2$$



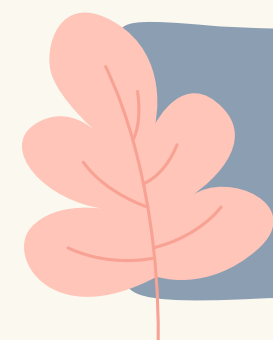


Quick Check



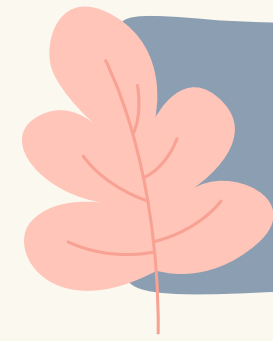
Question 1

$$\begin{array}{c} \text{6} \\ \text{+} \\ \text{6} \\ \text{+} \\ \text{6} \\ \text{=} \\ \text{?} \end{array}$$



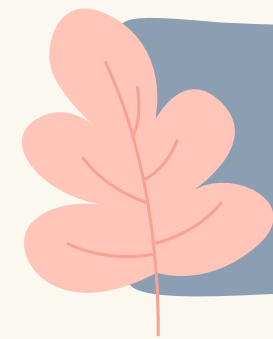
Answer 1

$$\begin{array}{c} \text{6} \\ \text{+} \\ \text{6} \\ \text{+} \\ \text{6} \\ \text{=} \\ \text{18} \end{array}$$



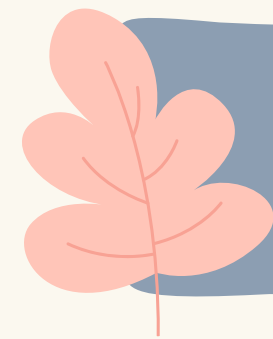
Question 2

$$\begin{array}{c} \text{4} \\ \text{+} \\ \text{4} \\ \text{+} \\ \text{4} \\ \text{+} \\ \text{4} \\ \text{=} \\ \text{?} \end{array}$$



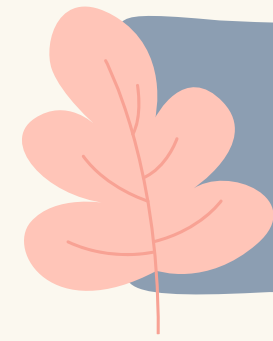
Answer 2

$$\begin{array}{c} \text{4} \\ \text{+} \\ \text{4} \\ \text{+} \\ \text{4} \\ \text{+} \\ \text{4} \\ \text{=} \\ \text{16} \end{array}$$



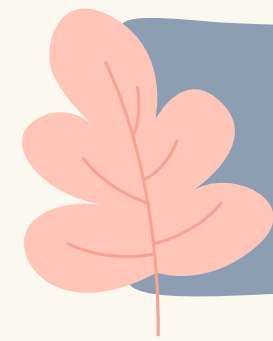
Question 3

$$3 + 3 + 3 + 3 = ?$$



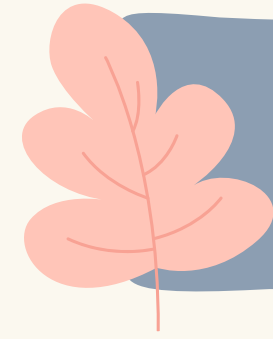
Answer 3

$$\textcircled{3} + \textcircled{3} + \textcircled{3} + \textcircled{3} = \textcircled{12}$$



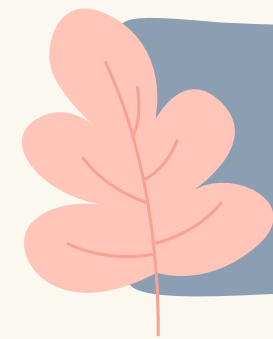
Question 4

$$5 + 5 + 5 = ?$$



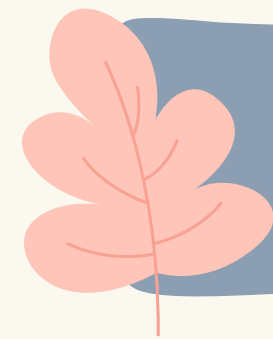
Answer 4

$$5 + 5 + 5 = 15$$



Question 5

$$\begin{array}{c} \text{2} \\ \text{+} \\ \text{2} \\ \text{+} \\ \text{2} \\ \text{+} \\ \text{2} \\ \text{+} \\ \text{2} \\ \text{=} \\ \text{?} \end{array}$$



Answer 5

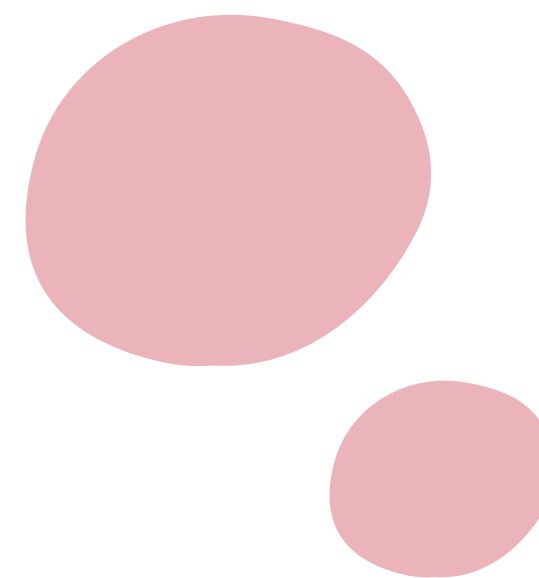
$$2 + 2 + 2 + 2 + 2 = 10$$



**Great
Job**



Multiplication Strategies

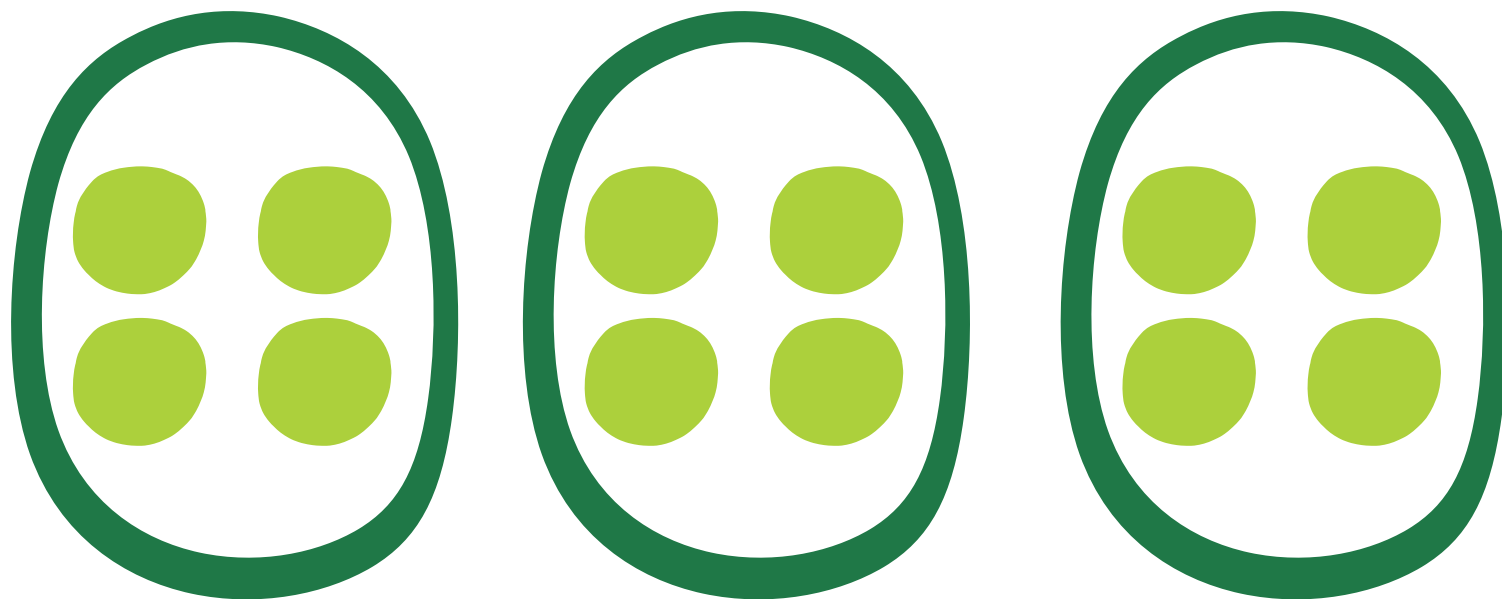


Equal Groups

Groups with an equal number of items in each group.

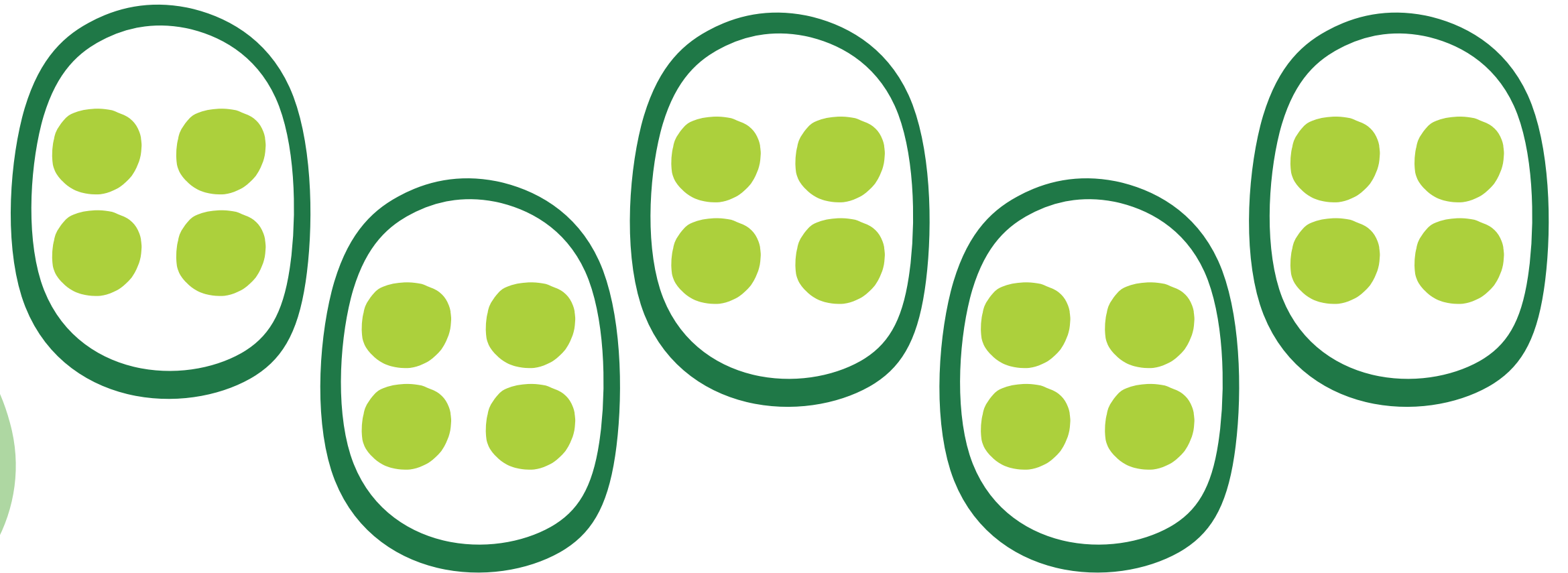
3 groups of 4

$$3 \times 4 = 12$$



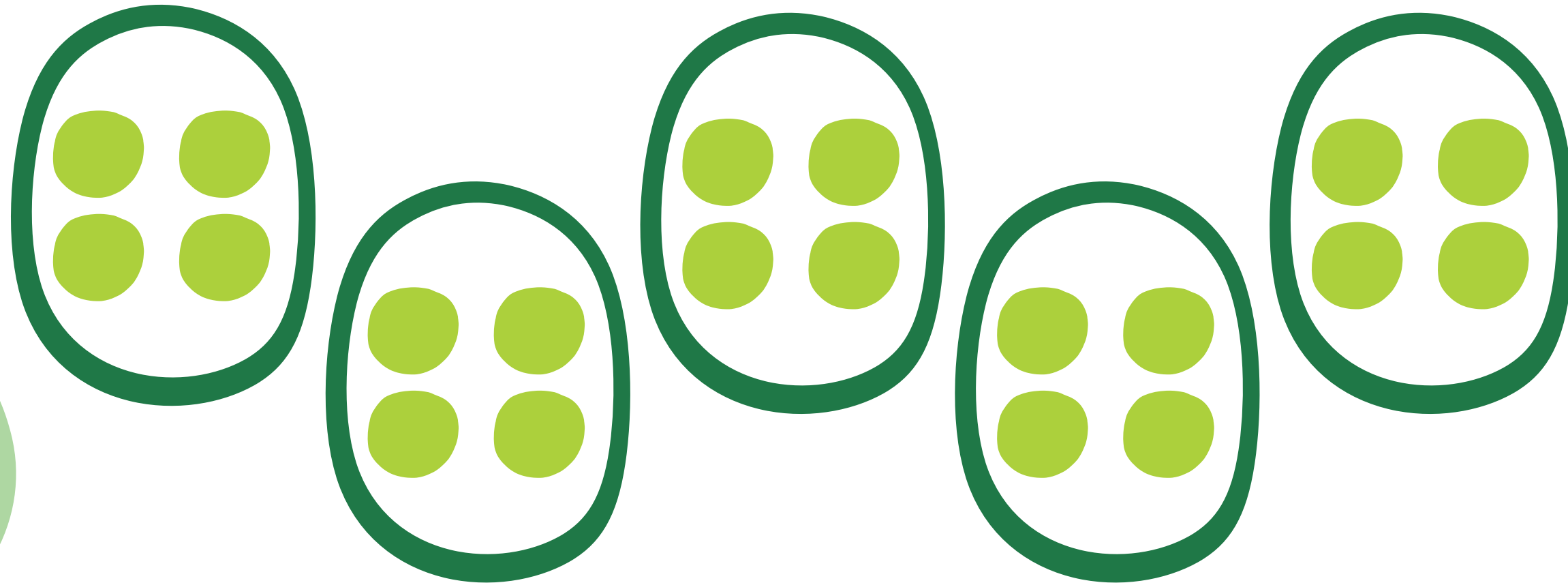
You
try!

Write a multiplication problem
to match the model:



Great!

5 groups of 4
 $5 \times 4 = 20$

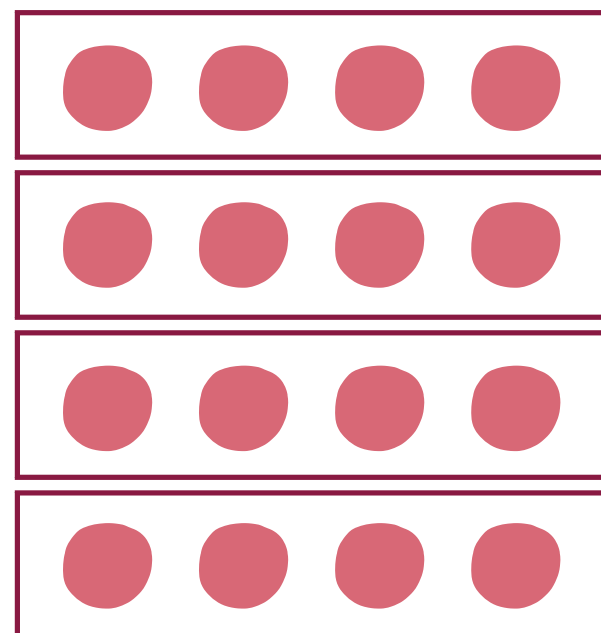


Array

Rows with an equal number of items in each row.

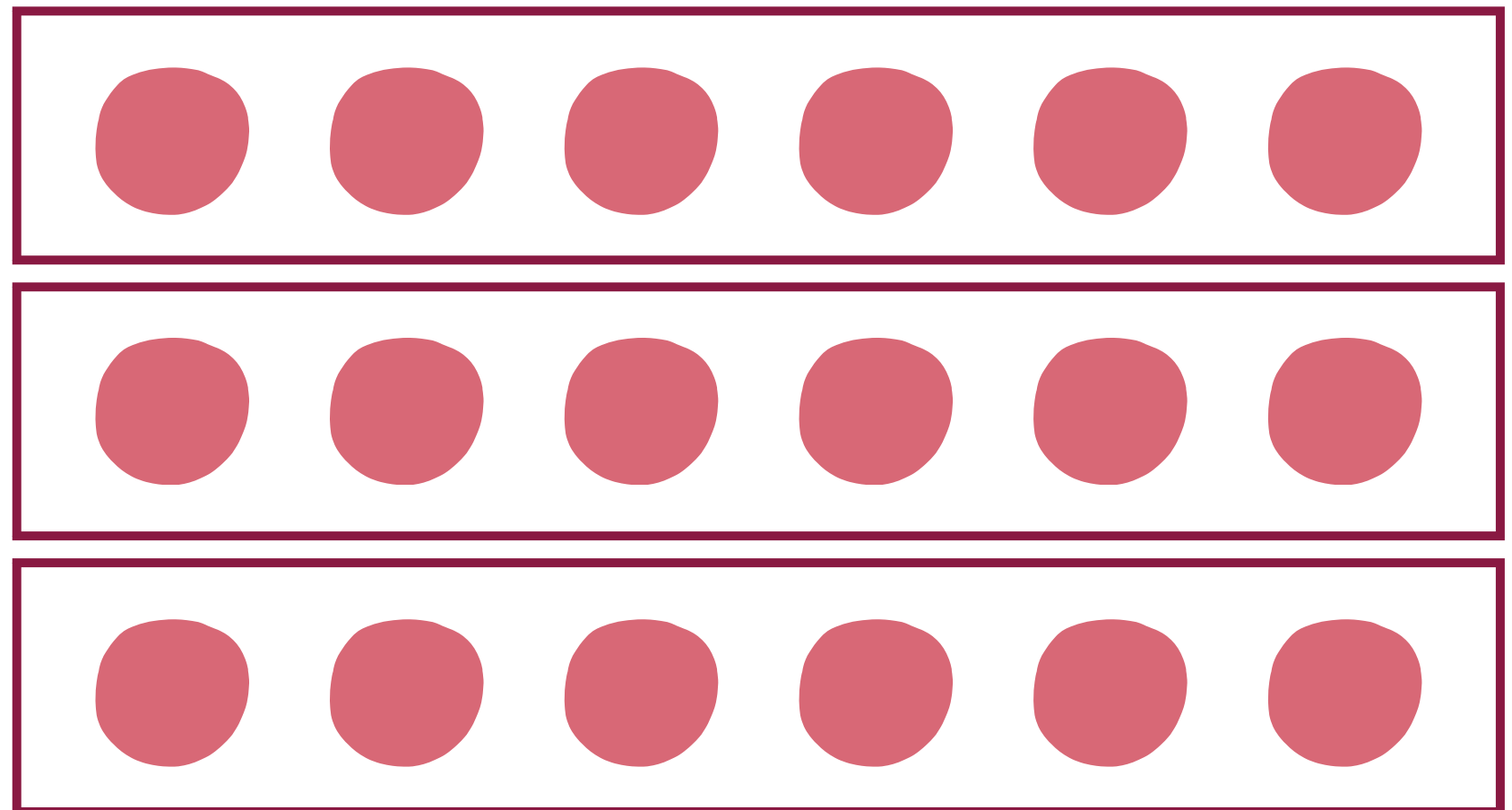
4 rows of 4

$$4 \times 4 = 16$$



You
try!

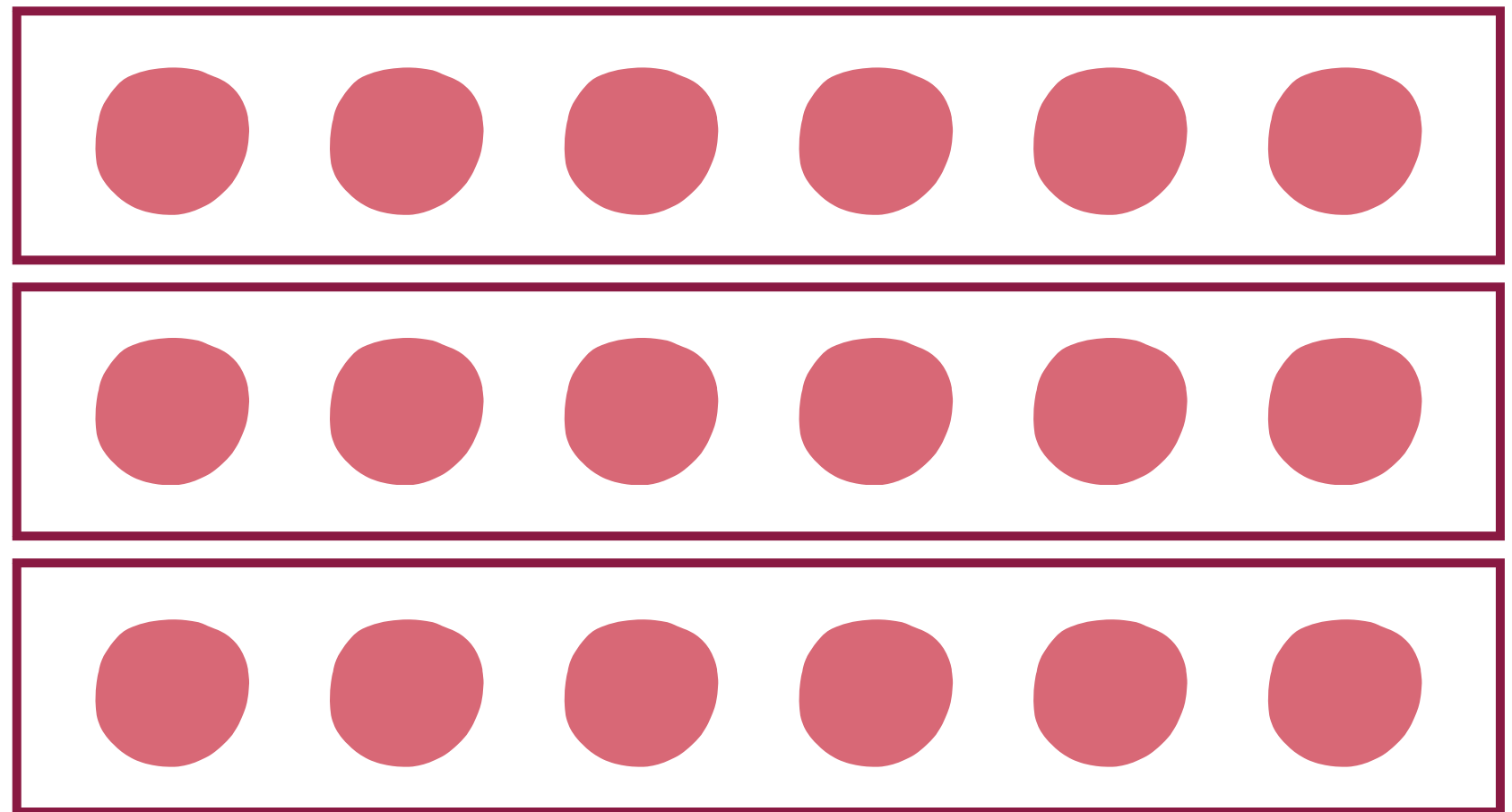
Write a multiplication problem
to match the model:



Awesome!

3 rows of 6

$$3 \times 6 = 18$$

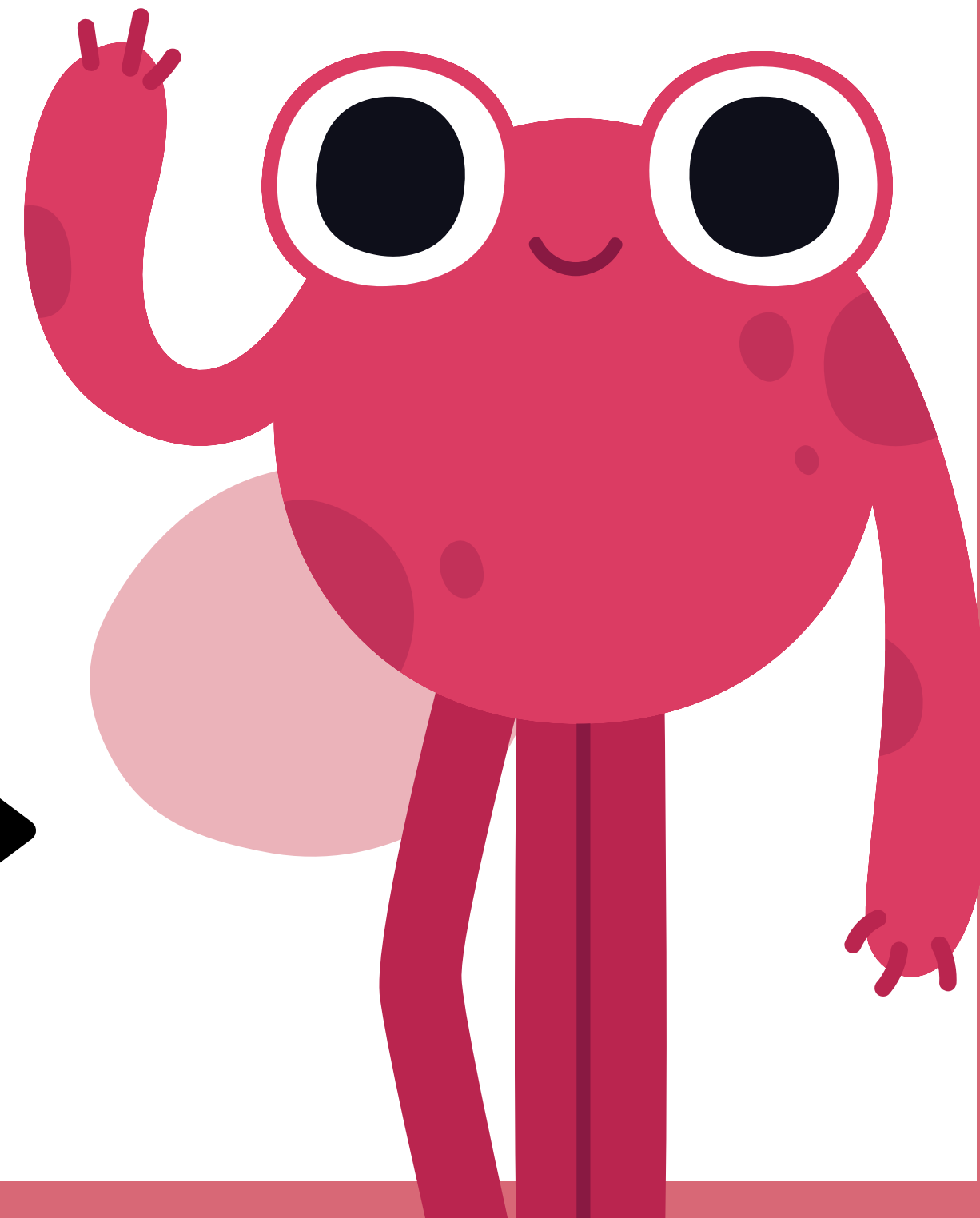
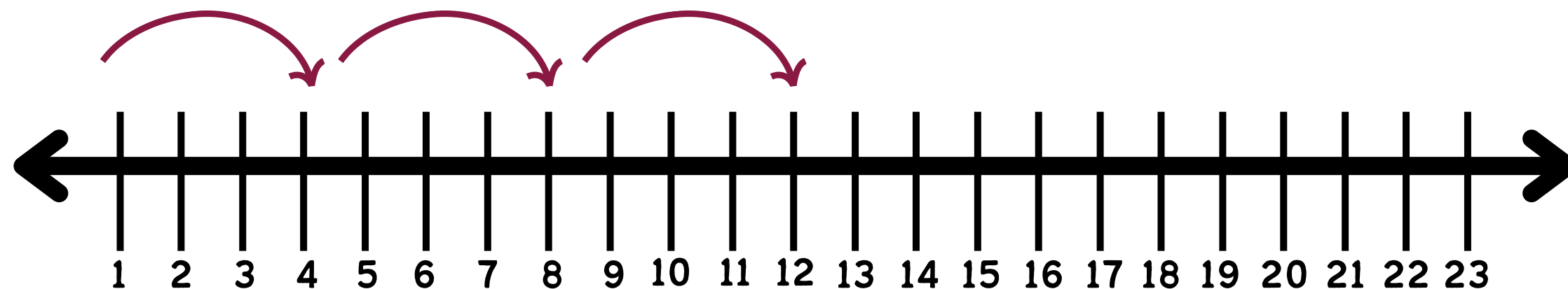


Number Line

Repeatedly skip the same amount on a number line.

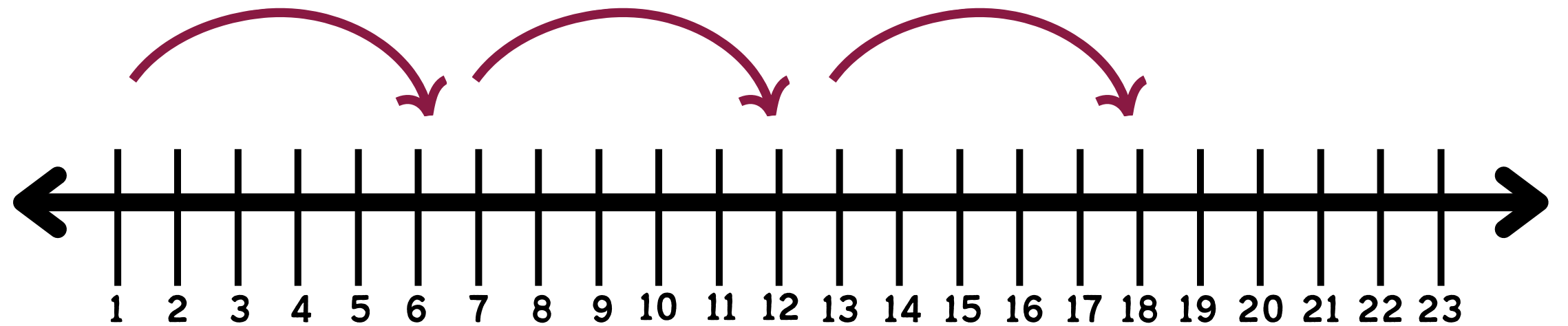
3 skips of 4

$$3 \times 4 = 12$$



You
try!

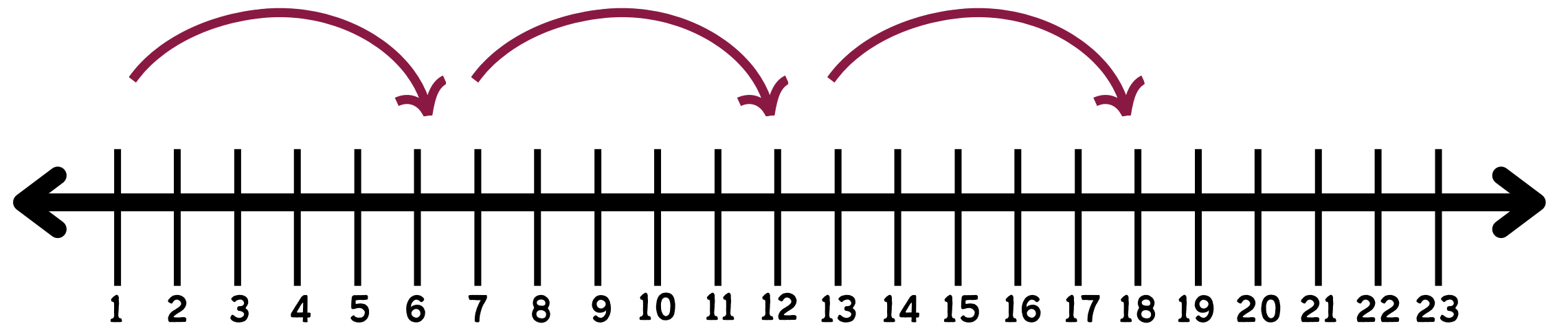
Write a multiplication problem
to match the model:



Perfect!

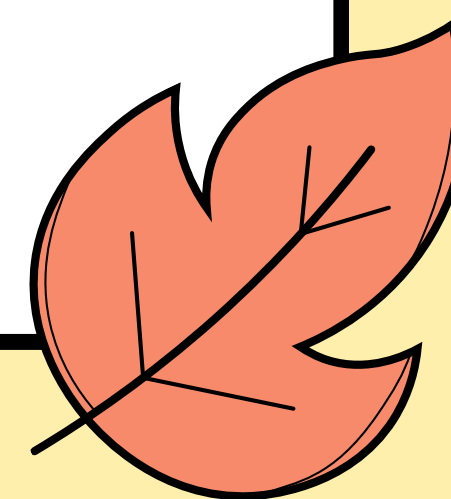
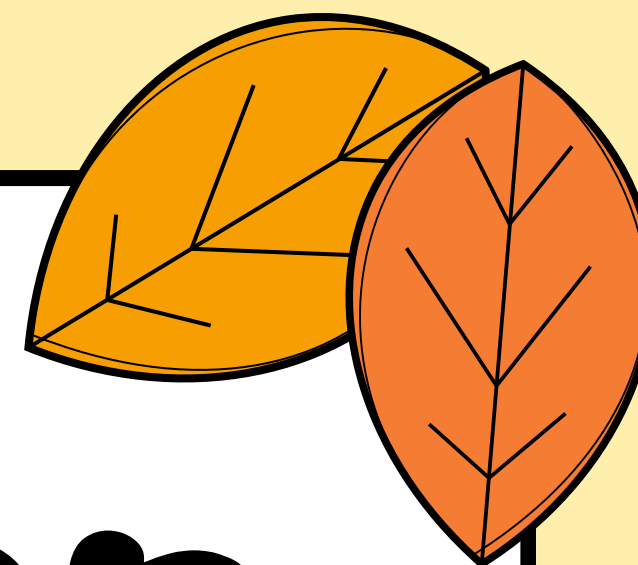
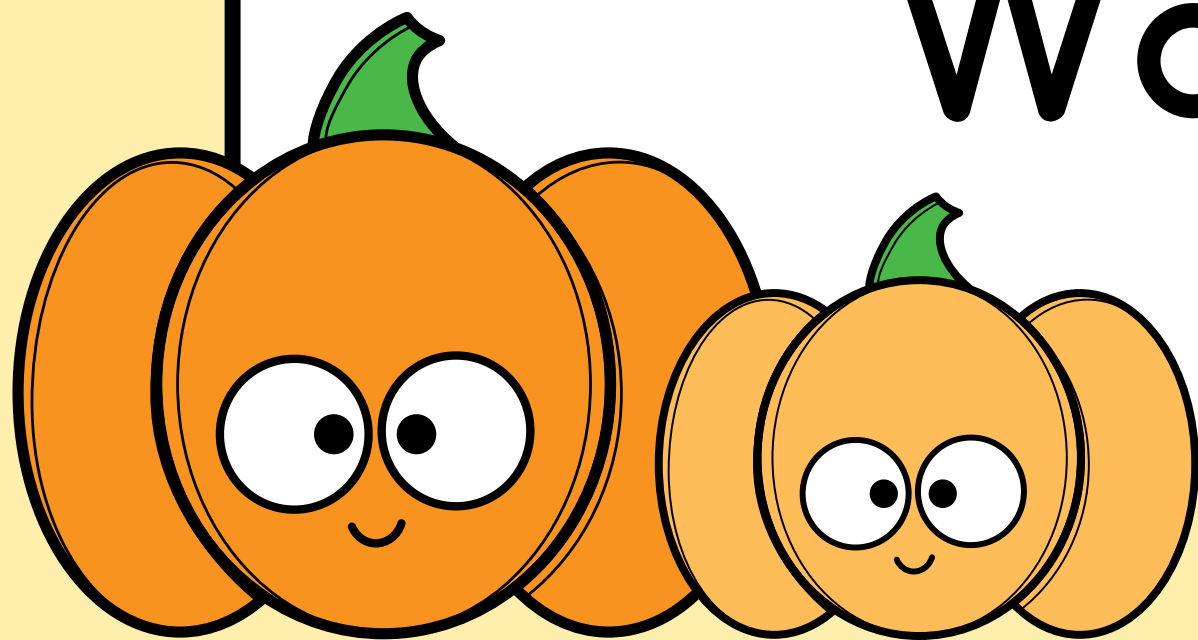
3 skips of 6

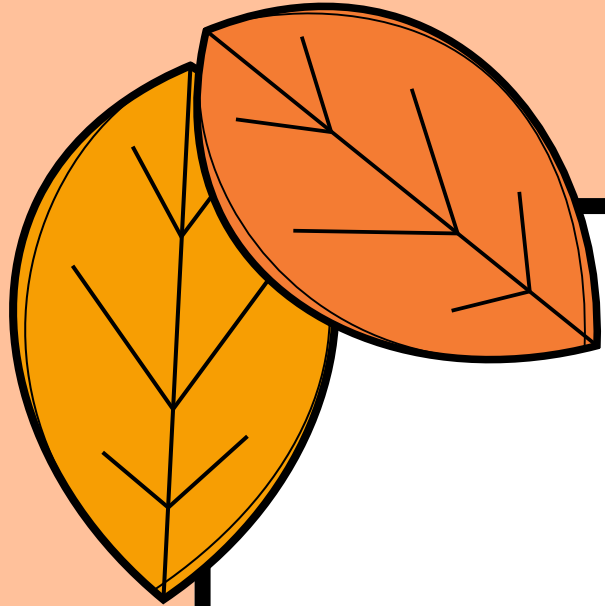
$$3 \times 6 = 18$$



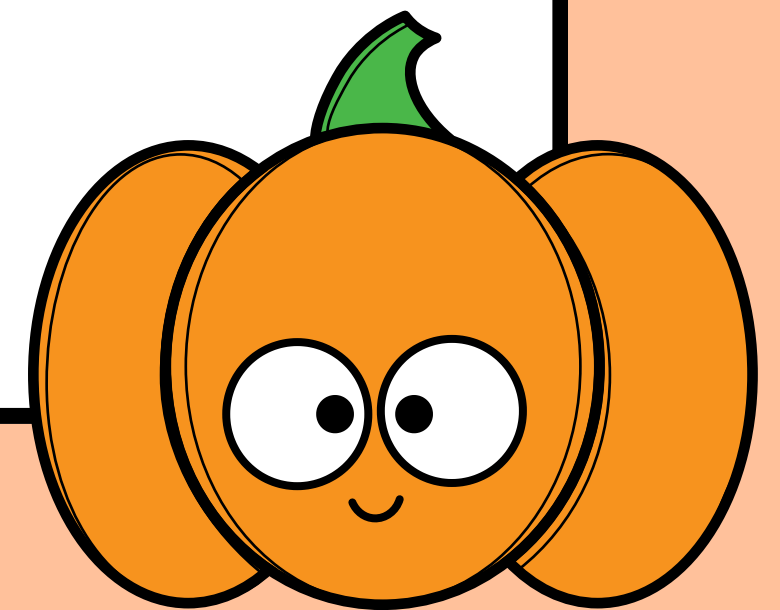
Multiplication

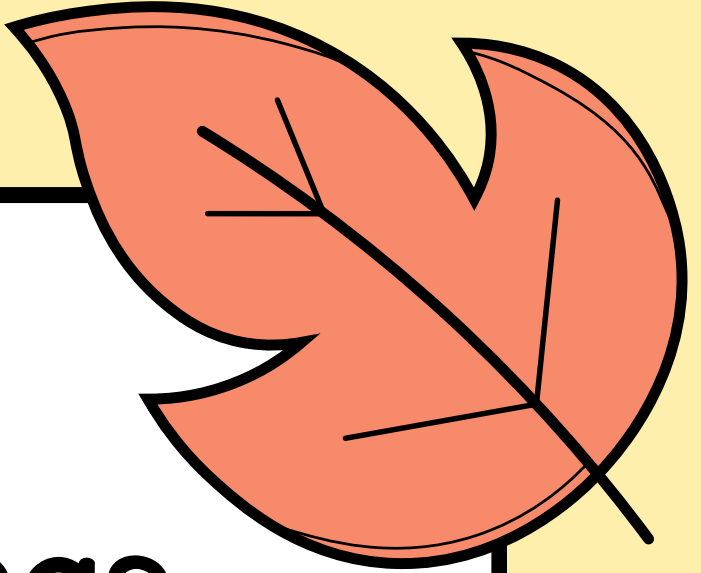
Word Problems



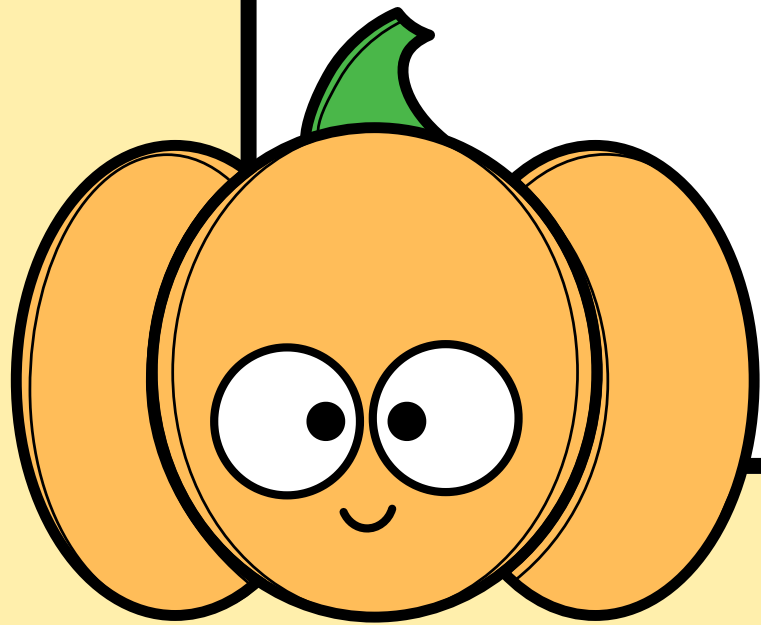


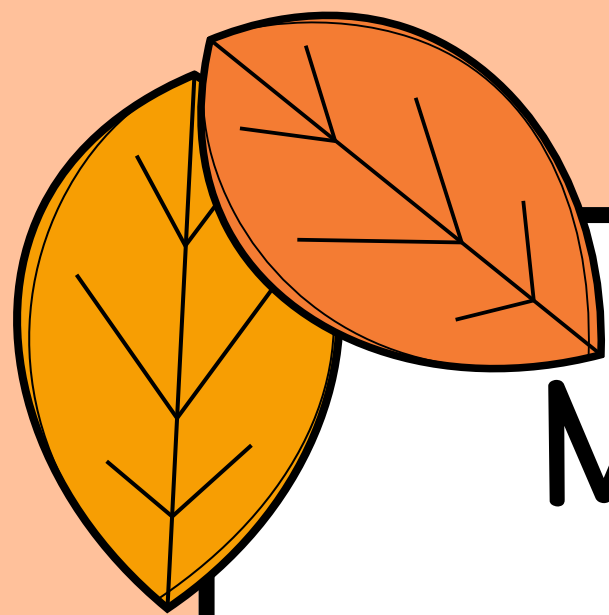
Josie has four wagons. There
are 10 pumpkins in each wagon.
How many pumpkins does she
have in all?



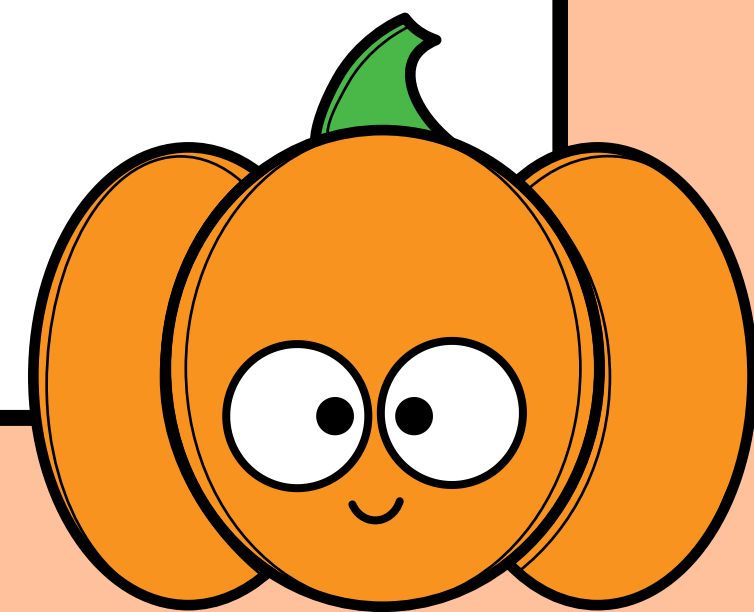


Caleb has two bags. Each bag has
12 acorns in it. How many acorns
does Caleb have in all?

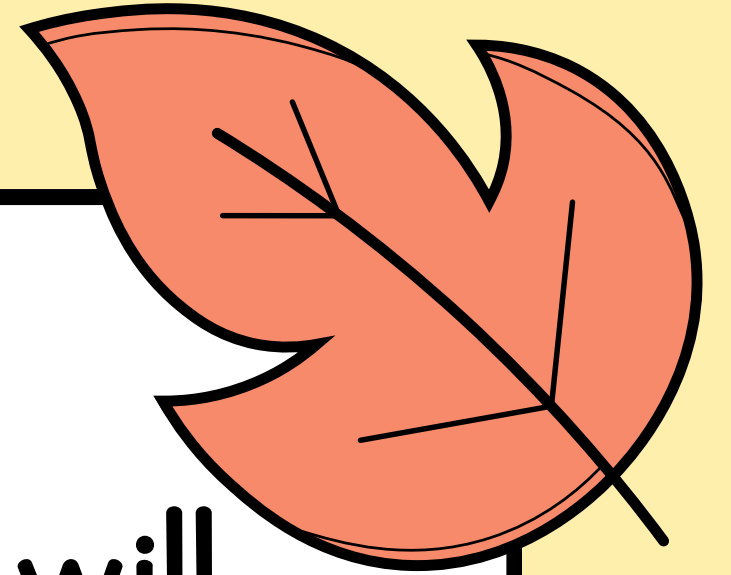
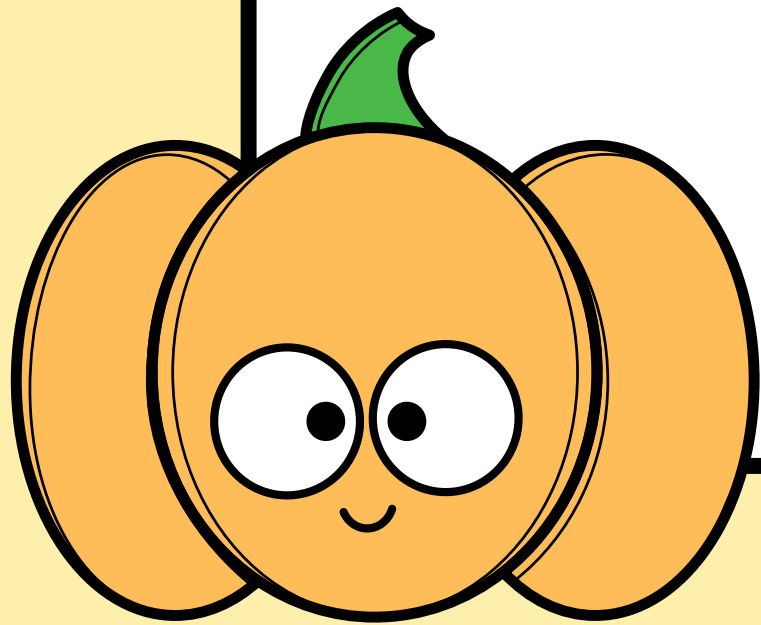


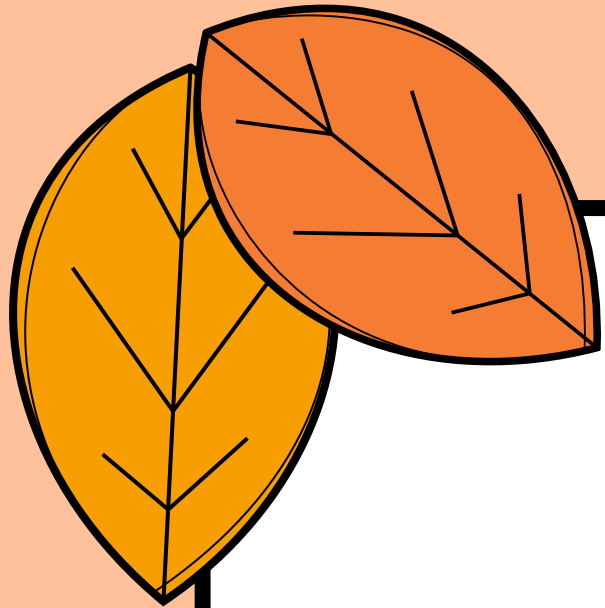


Megan's family has 8 bags full of
apples. Each bag contains 9 red
apples. How many apples do they
have in all?

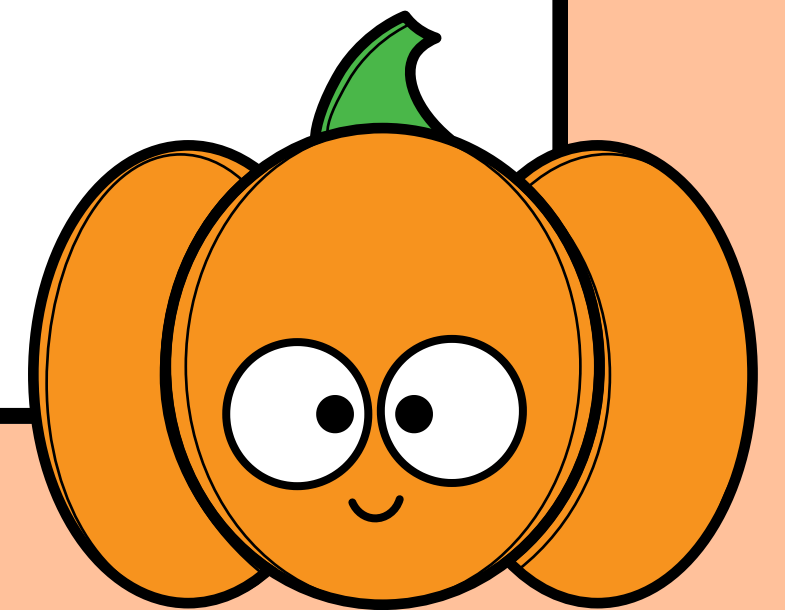


Max has 3 friends. Each friend will
get 5 cookies. How many cookies
will Max need in all?

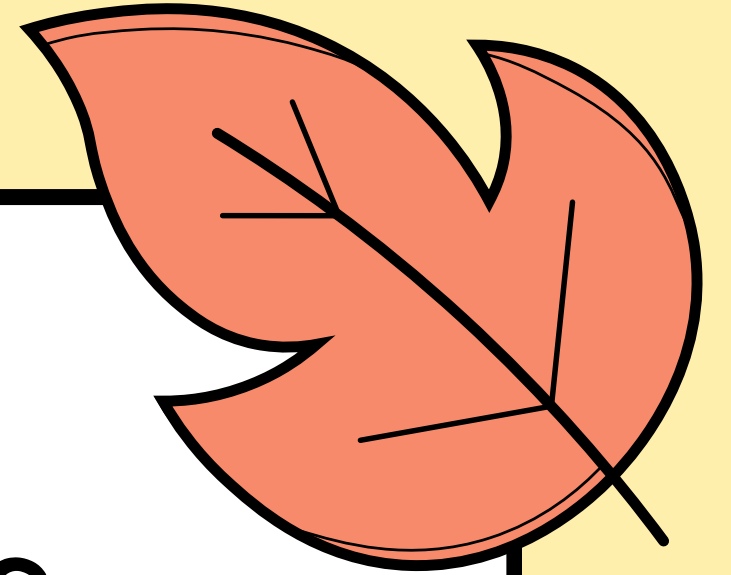
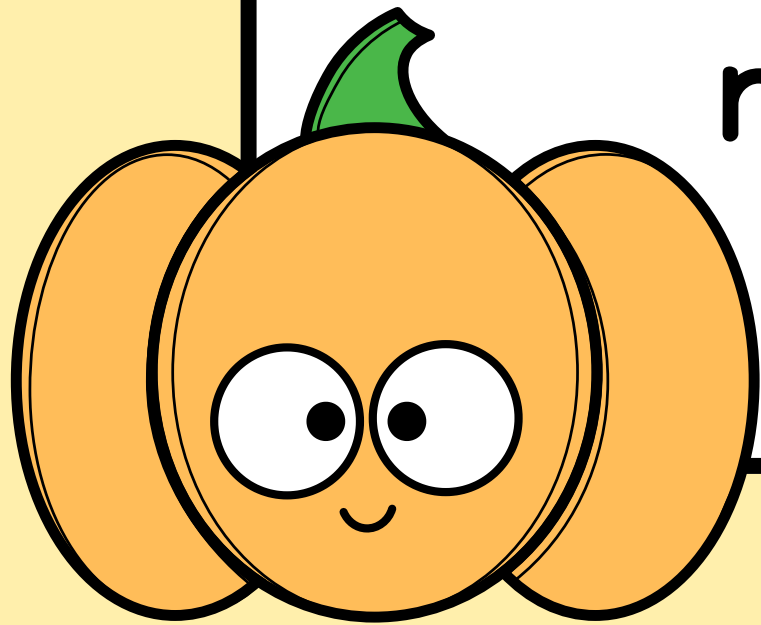


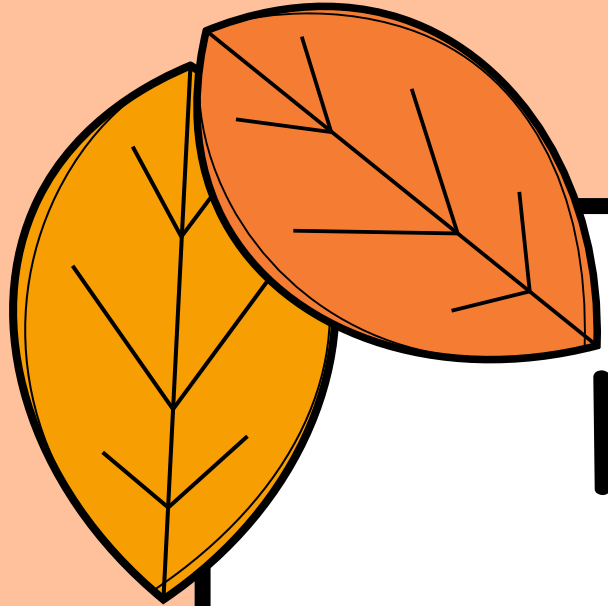


Ally has 2 pumpkins. Each
pumpkin has 11 seeds inside. How
many pumpkin seeds does Ally
have in total?

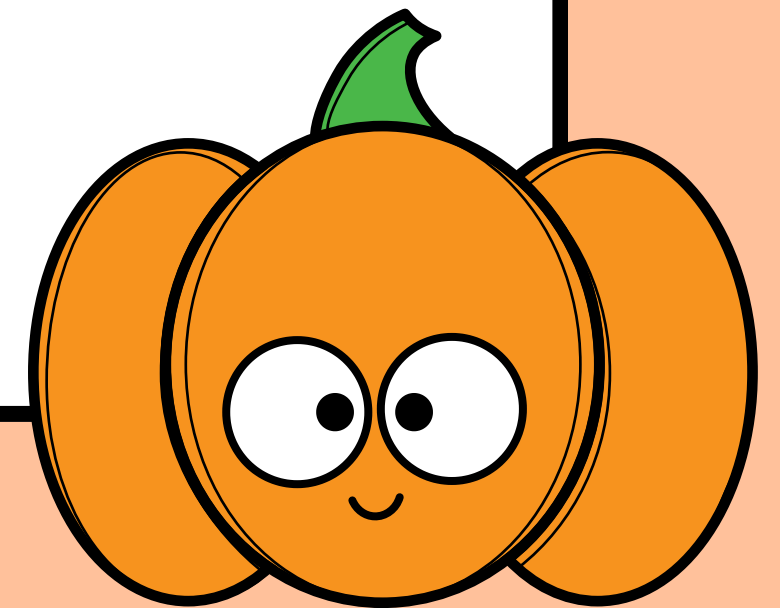


Harry has 7 buckets of sticks.
Each bucket has 8 sticks. How
many stick did Harry collect?





Kasey has 4 bundles of flowers.
Each bundle has 9 flowers. How
many flowers does Kasey have in
total?





THANK YOU

